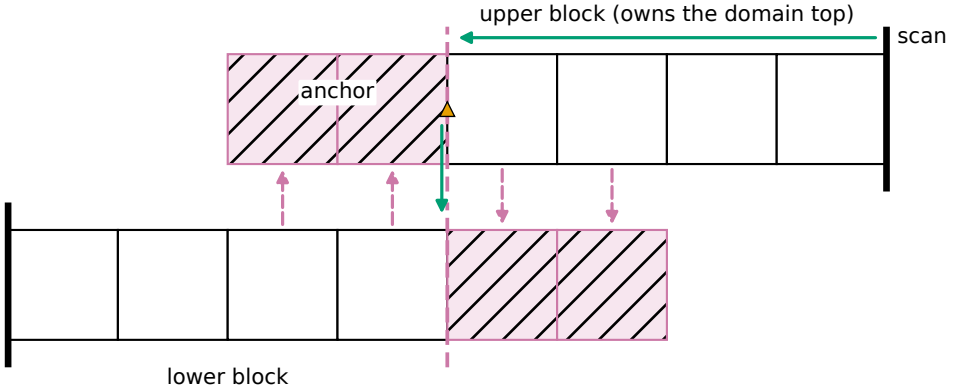

ghost rows of ρ_{ref}, ρ_{ref}
from the neighbour